Problem 5

A car moves along a horizontal circular path with a radius of $R = 43$ m. The car has a constant tangential acceleration $a = 2$ m/s^2. The initial speed of the car is $v_0 = 36$ km/h. Find the time it takes for the car to complete the first lap.

Solution

First, convert the initial speed from km/h to m/s:

$$v_0 = 36 \text{ km/h} = 36 \times \frac{1000}{3600} \text{ m/s} = 10 \text{ m/s}.$$

The distance the car needs to travel to complete one full lap is the circumference of the circle:

$$s = 2\pi R = 2\pi \times 43 = 86\pi \text{ m} \approx 270.2 \text{ m}.$$

Since the car is accelerating with a constant tangential acceleration a, we use the kinematic equation:

$$s = v_0 t + \frac{1}{2} a t^2.$$

Substituting the known values:

$$270.2 = 10t + \frac{1}{2} \times 2 \times t^2.$$

This simplifies to:

$$270.2 = 10t + t^2.$$

Rewriting it as a quadratic equation:

$$t^2 + 10t - 270.2 = 0.$$

Solve this using the quadratic formula:

$$t = \frac{-10 \pm \sqrt{10^2 - 4 \times 1 \times (-270.2)}}{2 \times 1}.$$

$$t = \frac{-10 \pm \sqrt{100 + 1080.8}}{2}.$$

$$t = \frac{-10 \pm \sqrt{1180.8}}{2}.$$

$$t = \frac{-10 \pm 34.36}{2}.$$

Taking the positive root:

$$t = \frac{-10 + 34.36}{2} = \frac{24.36}{2} = 12.18 \text{ s}.$$

Final Answer

The car takes approximately 12.18 s to complete the first lap.